

Inspect and Clean the Data

Project: Consumer Strategy, Food and Agriculture — CPG Global Entry Strategy

Student Name: Benjamin (Ru Yi) Cai (betakappa123) | **Team:** Wendy S (WendyS-28)

Which branch did you push to GitHub? cleaning-ben

Branch URL: <https://github.com/betakappa123/cpg-global-entry-strategy/tree/cleaning-ben>

Commit ID: afdfe60bf84efeae7688d0dd1fce84483c9c244

Primary Execution Notebook: DATA/DataCleaner.ipynb | **Clean Output Directory:** DATA/clean/

1. Cleaning Decision Log (Rule, Reason, Check)

ID	Cleaning Rule	Rationale / Reason	Verification Check Applied
R1	Header & Footer Boundary Isolation	Euromonitor exports contain title blocks in rows 0-4 and multi-row research/copyright footers that corrupt table parsing if ingested blindly.	Dynamically detect Geography header at row 5; terminate ingest at footer boundary ('Research Sources:'). Verified data row counts: 72, 864, 530, 7209.
R2	Non-Destructive Source Ingestion	Raw source files must remain 100% unchanged, reproducible, and verifiable.	Ingest raw .xls files in read-only mode via xlrld; execute cleaning in-memory; export clean files to separate DATA/clean/ directory.
R3	Text Standardization & Whitespace Cleaning	Pack type.xls (9 rows) and Pack Size.xls (56 rows) had 'PET Jars ' with trailing whitespace, breaking equality joins.	Strip leading/trailing whitespace across all text columns. Standardized 'PET Jars ' -> 'PET Jars'. Post-cleaning whitespace audit confirmed 0 remaining anomalies.
R4	Selective Hyphen / Missing Data Treatment	Text hyphens '-' represent two distinct realities: (a) genuinely unobserved markets (India RTD), vs (b) non-existent/inactive retail channels.	In Retail channels, hyphens treated as 0.0% volume share. In Market Sizes, India RTD hyphens kept strictly as NaN to prevent false zero consumption reporting.
R5	Semantic Duplicate Total Removal	In Retail channels.xls, both Outlet Type == 'Retail Channels' and Outlet Type == 'Total' report identical values (100.0% across all years), causing potential double counting.	Confirmed max absolute difference between 'Retail Channels' and 'Total' is 0.0 across all 385 comparisons. Dropped redundant 'Total' rows (396 tidy rows) from analytical merged files.
R6	Currency Scaling & FX Inconsistency Protection	In Market Sizes.xls, regional aggregates are in USD million, but countries are in local currencies, and Japan is reported in JPY billion (a 1,000x scale difference).	Flagged that cross-country dollar comparisons require external FX conversion. Preserved exact currency labels and units; used physical Market Volume (Tonnes / Litres) as primary cross-country comparison metric.
R7	Tidy Long-Format Reshaping	Raw wide formats (years as columns 2015–2025) impede multi-table joins, panel filtering, and regression/visualization.	Reshaped tables using pd.melt() into tidy long format: Year cast to int64, values cast to float64. Zero failed conversions.
R8	Multi-Table Relational Merge & Metric Derivation	Analyzing channel strategy requires knowing physical volume sold through each channel, not merely percentage shares.	Merged Market Sizes (Volume) with Retail Channels (% Share) on ['Geography', 'Category', 'Year']. Computed Channel_Volume = Market_Volume * (Channel_Share_Pct / 100). Exactly 9,108 rows; 0 unmatched keys.

2. Explanation of How One Codex / AI Suggestion Was Checked

Context: During exploratory automated script generation, an AI coding assistant (Codex/LLM) suggested a blanket imputation snippet: `df = df.replace('-', 0).infer_objects(copy=False)` to streamline numeric conversion and avoid

handling missing values.

Audit & Verification Applied: We inspected which exact rows in each dataset contained hyphens. In `Market Sizes.xls`, hyphens appeared exclusively in India RTD Coffee Off-trade Volume and Off-trade Value RSP across all 11 years (2015–2025). Consulting Euromonitor's research methodologies revealed that hyphens signify unobserved or untracked market segments (Euromonitor did not measure RTD liquid coffee in India during this horizon), *not* zero consumption.

Downstream Mathematical Fallout: Imputing India RTD volume as `0.0` would falsely imply zero consumption, cause divide-by-zero errors in CAGR calculations, and artificially dilute Asia-Pacific per-capita benchmarks across India's 1.4 billion population.

Decision Made: We **rejected** the uniform zero-imputation rule. In `Retail channels`, where hyphens represent inactive/unmeasured channels (e.g. forecourt retailers in China), values are treated as `0.0%`; in `Market Sizes`, hyphens are preserved strictly as `NaN` so unobserved market segments are excluded from aggregates without corrupting averages.

3. The 14-Item DSO 576 Data Cleaning Checklist

1 State the project vision and question

State your project vision and one business question. Name the main metric or comparison.

Status: Done

Evidence or explanation: Documented in `project-start.md` (lines 5–45), `dashboard.html` (lines 7–10), and Section 1 of `DATA/Datacleaner.ipynb`.

- **Project Vision:** Evaluate whether a Chinese coffee enterprise originating from Yunnan Province should expand into international overseas markets, identifying the highest-potential target markets in Asia-Pacific and globally to support strategic entry, category selection, and channel deployment.
- **One Business Question:** *"Which international markets and categories (packaged coffee vs. ready-to-drink [RTD] coffee) offer the highest sustainable volume growth and accessible retail/channel entry routes for a Yunnan coffee enterprise looking to expand overseas?"*
- **Main Metric or Comparison:** Market volume (Tonnes for Packaged Coffee, Million Litres for RTD Coffee) and 10-year CAGR (2015–2025), evaluated alongside off-trade retail channel shares (% E-commerce, Supermarkets, Hypermarkets, Convenience) and packaging unit volume mix (million units by pack format).

2 Define one row

What does one row represent? Name the column or combination that should uniquely identify a row, if any.

Status: **Done**

Evidence or explanation: Programmatically verified in Section 2 of `Datacleaner.ipynb` (zero duplicate candidate keys across all 4 datasets):

1. **Market Sizes.xls** : In wide format, one row represents an 11-year time series (2015–2025) of category volume or value for a unique Geography × Category × Data Type × Unit × Current Constant. In tidy long format, one row represents a single annual volume or retail value observation for a given Geography, Category, Data Type, and Year. Unique Key: `['Geography', 'Category', 'Data Type', 'Year']`.
2. **Retail channels.xls** : In wide format, one row represents the annual percentage distribution of sales volume across 2015–2025 for a specific Geography, Category, Outlet Type, and Data Type. In tidy format, one row represents the percentage share of volume sold through one Outlet Type in a given Geography, Category, and Year. Unique Key: `['Geography', 'Category', 'Outlet Type', 'Year']`.
3. **Pack type.xls** : In wide format, one row represents annual unit volume (million units) for a specific packaging container format (Pack Type) within a Geography, Category, and Packaging Class ('Total'). In tidy format, one row represents annual unit volume of a specific Pack Type in a given Geography, Category, and Year. Unique Key: `['Geography', 'Category', 'Packaging Class', 'Pack Type', 'Year']`.
4. **Pack Size.xls** : In wide format, one row represents annual unit volume (million units) for a granular pack size/capacity (Pack Size, e.g. '100 g', '250 ml', or 'Total') nested within a Pack Type, Packaging Class, Category, and Geography. In tidy format, one row represents annual unit volume of a specific Pack Size in a given Geography, Category, Pack Type, and Year. Unique Key: `['Geography', 'Category', 'Packaging Class', 'Pack Type', 'Pack Size', 'Year']`.

3 Preserve the source

List the data source and version or download date. Keep the original files unchanged. Explain how to load them.

Status: **Done**

Evidence or explanation: Documented in sheet footers and implemented via `inspect_and_load_source()` in Section 3 of `Datacleaner.ipynb`.

- **Data Source:** Euromonitor International via USC Libraries Passport database subscription (Hot Drinks, Soft Drinks, and Beverage Packaging modules).
- **Export Timestamps & Provenance:** Market Sizes.xls (GMT 9/20/2026 11:17:37 PM, 72 data rows); Retail channels.xls (GMT 10/1/2026 7:02:34 AM, 864 data rows); Pack type.xls (GMT 10/1/2026 7:05:16 AM, 530 data rows); Pack Size.xls (GMT 10/1/2026 7:07:44 AM, 7,209 data rows).
- **Preservation:** Original binary .xls files in `DATA/` remain strictly read-only and untouched.
- **Loading Procedure:** Loaded via `inspect_and_load_source()`. The parser dynamically locates the header row (Geography at row index 5), parses data rows up to the footer boundary, and isolates clean tables in-memory before exporting them to separate CSV files in `DATA/clean/`.

4 Inspect columns and types

List incorrect types and your conversions. Count failed conversions and explain how you handled them.

Status: **Done**

Evidence or explanation: Programmatically audited in Section 4 of `Datacleaner.ipynb`.

- **Conversions:** Year columns (2015–2025) imported as string object due to text hyphens ('-'). Converted to numeric float64 via `pd.to_numeric()`, mapping hyphens to `np.nan` (or 0.0 for inactive retail channels). In tidy format, converted Year from string to `int64`. Categorical columns typed as clean strings.
- **Failed Conversions Count: 0 unexpected failures.** Every non-numeric string across all four datasets was strictly the hyphen character '-'. Hyphen counts: Market Sizes = 22; Retail channels = 3,384; Pack type = 96; Pack Size = 2,758.

5 Standardize text

Show text before and after cleaning. Keep meaningful differences and leading zeros in IDs.

Status: **Done**

Evidence or explanation: Programmatically audited in Section 5 of `Datacleaner.ipynb`.

- **Text Cleaning:** In `Pack type.xls` (9 rows) and `Pack Size.xls` (56 rows), Pack Type contained an accidental trailing space: Before: 'PET Jars ' (length 9) → After: 'PET Jars' (length 8). Applied `.str.strip()` across all text columns across all tables. Trailing/leading whitespace issues after cleaning: **0**.
- **Preserved Differences:** Preserved category distinctions (Coffee vs RTD Coffee), currency scale distinctions (USD million, JPY billion, CNY million, GBP million, EUR million), and package units (Tonnes, million litres, million units, g, ml).
- **IDs:** Euromonitor uses standardized text strings rather than numeric codes with leading zeros.

6 Investigate duplicates

Count exact duplicate rows and repeated IDs separately. Explain which you kept or removed, and why.

Status: **Done**

Evidence or explanation: Programmatically verified in Section 6 of `Datacleaner.ipynb`.

- **Exact Duplicate Rows:** Market Sizes = 0; Retail channels = 0; Pack type = 0; Pack Size = 0.
- **Repeated IDs: 0 repeated composite primary keys** across all four tables in raw format.
- **Semantic Duplicates Handled:** (1) In `Retail channels.xls`, Outlet Type == 'Retail Channels' and Outlet Type == 'Total' both report identical values (100.0% across all years; max diff = 0.0). Retained 'Retail Channels' as primary total and filtered out redundant 'Total' rows (36 rows removed) to prevent double counting. (2) In `Pack Size.xls`, the 530 rows where Pack Size == 'Total' are exact replicas of `Pack type.xls`.

7 Handle missing data

Report missing counts before and after cleaning for each affected column. Explain why you kept, filled, or dropped values.

Status: **Done**

Evidence or explanation: Programmatically verified in Section 7 of `Datacleaner.ipynb`.

- **Market Sizes.xls (22 missing values):** India RTD Coffee Off-trade Volume and Value across 11 years. Kept strictly as `NaN` (NOT filled with 0) to avoid falsely reporting zero consumption and distorting growth metrics.
- **Retail channels.xls (3,384 missing values, 35.61% of channel cells):** Inactive/unmeasured retail channels in specific countries (e.g. forecourt retailers in China). Treated as 0.0% for additive channel share calculations; preserved as `NaN` for presence analysis.
- **Pack type.xls (96 missing values) & Pack Size.xls (2,758 missing values):** Preserved as `NaN` when analyzing launch timelines; treated as 0.0 when computing aggregate packaging volume.

8 Check suspicious values

Check invalid dates, impossible values, and unusual records. Explain what you corrected, kept, or flagged. Unusual does not always mean wrong.

Status: **Done**

Evidence or explanation: Programmatically verified in Section 8 of `Datacleaner.ipynb`.

- **Date Bounds:** All years span 2015 to 2025. Historical: 2015–2023; Forecast: 2024–2025. Valid range.
- **Negative Values: 0 negative values** across all four datasets (minimum numeric value is 0.0).
- **Percentages:** All channel shares in Retail channels fall strictly within [0.0, 100.0]. Total channel sums equal 100.0%.
- **CRITICAL FLAG — Currency Scaling Discrepancies:** Regional aggregates report Retail Value RSP in USD million, whereas country-level rows report retail value in local currency (China: CNY million; India: INR million; Japan: JPY billion [1,000x scale difference!]; Thailand: THB million; Australia: AUD million; Brazil: BRL million; France/Germany: EUR million; UK: GBP million; USA: USD million).
- **Correction / Action:** Direct dollar comparisons across countries cannot be made from raw Retail Value RSP without external exchange rates. Market Volume (Tonnes for coffee, Million Litres for RTD) must be used as primary physically consistent metric.

9 Verify merges

If you merged tables, list the keys and expected relationship. Report unmatched keys, row counts before and after, and any unexpected extra rows.

Status: **Done**

Evidence or explanation: Programmatically verified in Section 9 of `Datacleaner.ipynb`.

- **Primary Merge:** Market Sizes (Volume) merged with Retail Channels (% Share) on `['Geography', 'Category', 'Year']`. Relationship: One-to-Many. Unmatched Keys: 0 on either side (18 geographies × 2 categories × 11 years = 396 groups perfectly matched). Pre-merge: Market Sizes Volume tidy = 396; Retail Channels tidy (filtered) = 9,108. Post-merge: Exactly 9,108 rows (0 unexpected extra rows, 0 dropped rows). Derived Metric: $\text{Channel_Volume} = \text{Total_Market_Volume} * (\text{Channel_Share_Pct} / 100.0)$.
- **Secondary Merge Validation:** Pack type.xls merged with Pack Size.xls ('Total') on `['Geography', 'Category', 'Packaging Class', 'Pack Type Clean']`. Exactly 530 rows matched 1-to-1, 0 unmatched, 0 numeric discrepancies across all 11 years.

10 Reconcile changes

Report starting and final row counts. Account for the difference. Explain changes to important totals after cleaning or merging.

Status: **Done**

Evidence or explanation: Programmatically audited in Section 10 of `Datacleaner.ipynb`.

- **Row Counts Reconciliation:** Market Sizes: raw 85 → 72 data rows → 792 tidy rows (8 footer rows parsed). Retail Channels: raw 877 → 864 data rows → 9,504 tidy rows → 9,108 clean rows (36 redundant 'Total' rows removed; 8 footer rows). Pack Type: raw 542 → 530 data rows → 5,830 tidy rows (7 footer rows). Pack Size: raw 7,221 → 7,209 data rows → 79,299 tidy rows (7 footer rows).
- **Reconciliation of Totals:** Summing granular pack sizes (Pack Size != 'Total') within each pack type reproduces reported Pack Type totals within a mean difference of < 0.07 million units (max discrepancy < 0.8 million units across all years), which is mathematically attributable to 1-decimal rounding in raw source tables. Retail channel sub-shares sum cleanly to 100.0% within rounding tolerance.

11 Check calculations

Explain how missing values affect totals, averages, and denominators. Give the record count for each metric. State how you handle missing group keys.

Status: **Done**

Evidence or explanation: Programmatically verified in Section 11 of `Datacleaner.ipynb`.

- **Impact of Missing Values:** In RTD Coffee regional sums, India is omitted because its volume is unobserved (NaN). For channel penetration, including zeroes measures cross-market portfolio availability, while excluding zeroes measures average penetration among active markets.
- **Record Counts by Metric:** Market Sizes Volume: 396 observations (374 valid, 22 missing for India RTD); Market Sizes Value: 396 observations; Retail Channels (non-redundant): 9,108 observations (5,724 active non-zero, 3,384 inactive/0%); Pack Type: 5,830 observations; Pack Size: 79,299 observations.
- **Missing Group Keys:** 0 missing group keys across all datasets (no null or empty strings exist in any grouping dimension).

12 Verify individual records

Check five records, including difficult cases. Write the expected result and reason before checking the output. Show the original value, actual result, and whether it matches.

Status: **Done**

Evidence or explanation: Audited across five targeted cases with 100% agreement. Detailed audit table included in Section 4 below.

13 Test reproducibility

Restart and run all code from the original data using the listed dependencies. Give the command or notebook steps. Confirm that outputs match your submitted files.

Status: Done

Evidence or explanation: Tested and confirmed in Section 13 of `Datacleaner.ipynb`.

• **Dependencies:** Python 3.10+ (tested on Python 3.14.0), `pandas >= 2.0`, `xlrd >= 2.0.1`, `numpy >= 1.24`, `nbformat >= 5.11`, `nbclient >= 0.11`.

• **Reproducibility Command:** `python -m jupyter nbconvert --to notebook --execute Datacleaner.ipynb --output Datacleaner.ipynb`

• **Clean Output Directory:** Clean datasets exported to `DATA/clean/`: `clean_market_channels_merged.csv` (858 KB), `clean_market_sizes_tidy.csv` (23 KB), `clean_market_sizes_wide.csv` (10 KB), `clean_retail_channels_tidy.csv` (611 KB), `clean_retail_channels_wide.csv` (85 KB), `clean_pack_type_tidy.csv` (561 KB), `cleaned_sample_50.csv` (5 KB).

14 Document remaining limitations

List unresolved issues. Explain how they could affect your conclusions and what information you still need.

Status: Done

Evidence or explanation: Documented in Section 14 of `Datacleaner.ipynb` and `data-fit.md` (lines 79–84).

1. **Cross-Country Currency Comparability:** Country retail value is in local currency (CNY, INR, JPY, EUR, GBP, THB) without annual FX exchange rates. Direct revenue comparisons require linking an auxiliary exchange-rate table (e.g. IMF/World Bank annual FX rates). Strategic analysis must prioritize volume metrics (Tonnes and Litres).
2. **Missing India RTD Coffee Data:** India RTD coffee is unobserved in Passport. Market sizing for India must be restricted to Packaged Coffee.
3. **Exclusion of Foodservice / On-Premise Consumption:** Passport covers off-trade retail channels only. Out-of-home consumption (cafés, restaurants, specialty coffee shops) is unobserved.
4. **Rounding in Granular Pack Sizes:** Granular pack sizes have minor decimal rounding differences (< 0.8 million units) against reported totals.

4. Five Record Checks (Original, Expected, Actual)

Case	Original Value	Expected Result	Actual Result	Match?	Reason & Verification Scope
1. China Coffee Volume 2025	61314.5	61,314.5 Tonnes	61,314.5 Tonnes	Yes	Forecast home market volume benchmark
2. India RTD Coffee Volume 2020	'-' (Hyphen)	NaN (Unobserved)	NaN	Yes	Unobserved category preserved without false zero imputation
3. China Coffee E-Commerce 2025	46.7	46.7%	46.7%	Yes	Verifies digital channel transition (expanded from 10.8% in 2015)
4. PET Jars Text Standardization	'PET Jars '	'PET Jars'	'PET Jars'	Yes	Verifies elimination of trailing whitespace
5. Japan Coffee Value RSP 2023	415.5	415.5 JPY billion	415.5 JPY billion	Yes	Verifies currency unit and scale preservation (JPY billion)

5. Cleaned Data Sample (50 Representative Rows)

Sample Selection Rationale: The 50 records below are drawn from `clean_market_channels_merged.csv` to represent key strategic entry markets (China, Japan, Thailand, India, USA), both product categories (Coffee in Tonnes, RTD Coffee in Million

Litres), critical retail channels (E-Commerce, Supermarkets, Hypermarkets, Convenience Stores), and difficult edge cases (e.g. unobserved India RTD volume preserved as NaN).

#	Geography	Category	Outlet Type	Year	Share %	Total Mkt Vol	Unit	Channel Vol
1	China	Coffee	Convenience Stores	2015	5.4%	63,458.7	Tonnes	3,426.8
2	China	Coffee	Supermarkets	2015	31.6%	63,458.7	Tonnes	20,052.9
3	China	Coffee	Hypermarkets	2015	42.5%	63,458.7	Tonnes	26,969.9
4	China	Coffee	Retail E-Commerce	2015	10.8%	63,458.7	Tonnes	6,853.5
5	China	Coffee	Convenience Stores	2020	5.5%	67,782.7	Tonnes	3,728.0
6	China	Coffee	Supermarkets	2020	27.0%	67,782.7	Tonnes	18,301.3
7	China	Coffee	Hypermarkets	2020	33.9%	67,782.7	Tonnes	22,978.3
8	China	Coffee	Retail E-Commerce	2020	29.5%	67,782.7	Tonnes	19,995.9
9	China	Coffee	Convenience Stores	2025	5.5%	61,314.5	Tonnes	3,372.3
10	China	Coffee	Supermarkets	2025	20.5%	61,314.5	Tonnes	12,569.5
11	China	Coffee	Hypermarkets	2025	24.1%	61,314.5	Tonnes	14,776.8
12	China	Coffee	Retail E-Commerce	2025	46.7%	61,314.5	Tonnes	28,633.9
13	China	RTD Coffee	Convenience Stores	2015	6.5%	292.4	millio	19.0
14	China	RTD Coffee	Supermarkets	2015	36.6%	292.4	millio	107.0
15	China	RTD Coffee	Retail E-Commerce	2015	1.9%	292.4	millio	5.6
16	China	RTD Coffee	Convenience Stores	2020	14.5%	268.5	millio	38.9
17	China	RTD Coffee	Supermarkets	2020	19.1%	268.5	millio	51.3
18	China	RTD Coffee	Retail E-Commerce	2020	3.6%	268.5	millio	9.7
19	China	RTD Coffee	Convenience Stores	2025	16.7%	301.3	millio	50.3
20	China	RTD Coffee	Supermarkets	2025	18.9%	301.3	millio	56.9
21	China	RTD Coffee	Retail E-Commerce	2025	6.4%	301.3	millio	19.3
22	Japan	Coffee	Convenience Stores	2020	3.2%	119,096.3	Tonnes	3,811.1
23	Japan	Coffee	Supermarkets	2020	47.7%	119,096.3	Tonnes	56,808.9
24	Japan	Coffee	Retail E-Commerce	2020	9.6%	119,096.3	Tonnes	11,433.2
25	Japan	RTD Coffee	Convenience Stores	2020	21.4%	3,011.8	millio	644.5
26	Japan	RTD Coffee	Supermarkets	2020	24.5%	3,011.8	millio	737.9
27	Japan	RTD Coffee	Retail E-Commerce	2020	0.6%	3,011.8	millio	18.1
28	Japan	Coffee	Convenience Stores	2025	2.8%	97,775.2	Tonnes	2,737.7
29	Japan	Coffee	Supermarkets	2025	46.4%	97,775.2	Tonnes	45,367.7
30	Japan	Coffee	Retail E-Commerce	2025	12.4%	97,775.2	Tonnes	12,124.1
31	Japan	RTD Coffee	Convenience Stores	2025	21.8%	2,846.4	millio	620.5
32	Japan	RTD Coffee	Supermarkets	2025	23.2%	2,846.4	millio	660.4
33	Japan	RTD Coffee	Retail E-Commerce	2025	1.2%	2,846.4	millio	34.2
34	Thailand	Coffee	Convenience Stores	2020	16.3%	83,872.4	Tonnes	13,671.2
35	Thailand	Coffee	Supermarkets	2020	6.4%	83,872.4	Tonnes	5,367.8

36	Thailand	Coffee	Retail E-Commerce	2020	2.2%	83,872.4	Tonnes	1,845.2
37	Thailand	RTD Coffee	Convenience Stores	2020	14.7%	180.8	millio	26.6
38	Thailand	RTD Coffee	Supermarkets	2020	4.9%	180.8	millio	8.9
39	Thailand	RTD Coffee	Retail E-Commerce	2020	1.9%	180.8	millio	3.4
40	Thailand	Coffee	Convenience Stores	2025	16.3%	89,133.0	Tonnes	14,528.7
41	Thailand	Coffee	Supermarkets	2025	7.3%	89,133.0	Tonnes	6,506.7
42	Thailand	Coffee	Retail E-Commerce	2025	3.5%	89,133.0	Tonnes	3,119.7
43	Thailand	RTD Coffee	Convenience Stores	2025	19.0%	182.8	millio	34.7
44	Thailand	RTD Coffee	Supermarkets	2025	5.0%	182.8	millio	9.1
45	Thailand	RTD Coffee	Retail E-Commerce	2025	2.1%	182.8	millio	3.8
46	USA	Coffee	Supermarkets	2020	27.0%	858,536.5	Tonnes	231,804.9
47	USA	Coffee	Discounters	2020	2.1%	858,536.5	Tonnes	18,029.3
48	USA	Coffee	Retail E-Commerce	2020	17.9%	858,536.5	Tonnes	153,678.0
49	USA	Coffee	Supermarkets	2025	23.1%	735,187.2	Tonnes	169,828.2
50	USA	Coffee	Discounters	2025	1.8%	735,187.2	Tonnes	13,233.4